

DEEP LEARNING

The Complete Expert Curriculum

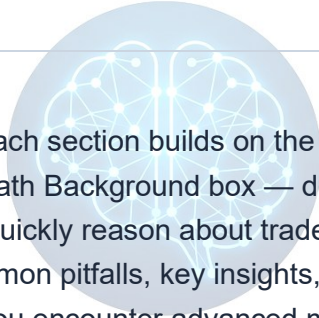
From Fundamentals to Production Mastery

What this document covers

This curriculum is structured to take you from zero assumptions to expert-level mastery of deep learning. Each topic is treated not as a definition to memorize but as a concept to deeply understand — covering the mathematical foundation, practical intuition, failure modes, performance characteristics, real-world use cases, and trade-offs against alternatives. You will finish not just knowing what techniques exist, but why they work, when to use them, when to avoid them, and how to reason about them from first principles.

How to use this guide

- Read sections in order — each section builds on the previous one.
- Every concept includes a Math Background box — do not skip it, even if approximate.
- Comparison tables let you quickly reason about trade-offs between methods.
- Callout boxes highlight common pitfalls, key insights, and practical tips.
- Revisit earlier sections as you encounter advanced material — depth accumulates.



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COMPLETE CURRICULUM OVERVIEW

Below is the full logical learning path. Every concept in this document is listed in the order you should encounter it.

Section 1 — Mathematical & Statistical Foundations

- Linear Algebra: vectors, matrices, tensors, operations
- Calculus: derivatives, chain rule, partial derivatives, gradients
- Probability & Statistics: distributions, Bayes' theorem, MLE
- Information Theory: entropy, cross-entropy, KL divergence

Section 2 — Core Neural Network Concepts

- The Perceptron and Biological Inspiration
- Feedforward Neural Networks (MLPs)
- Activation Functions: Sigmoid, Tanh, ReLU, Leaky ReLU, GELU, Swish
- Loss Functions: MSE, Cross-Entropy, Hinge, Focal, CTC
- Forward Pass & Computational Graphs
- Backpropagation Algorithm
- Vanishing & Exploding Gradients

Section 3 — Training Deep Networks

- Gradient Descent Variants: SGD, Momentum, Nesterov
- Adaptive Optimizers: AdaGrad, RMSProp, Adam, AdamW, LAMB
- Learning Rate Strategies: warm-up, decay, cosine annealing, cyclical
- Weight Initialization: Xavier, He, Orthogonal
- Normalization Techniques: Batch Norm, Layer Norm, Instance Norm, Group Norm, RMS Norm
- Regularization: L1/L2, Dropout, DropConnect, Stochastic Depth
- Data Augmentation Strategies
- Curriculum Learning & Training Stability

Section 4 — Convolutional Neural Networks

- Convolution Operation: filters, stride, padding, dilation
- Pooling: Max, Average, Global Average Pooling
- Classic Architectures: LeNet, AlexNet, VGG, Inception, ResNet, DenseNet
- Modern Efficient CNNs: EfficientNet, MobileNet, ShuffleNet
- Object Detection: R-CNN family, YOLO, SSD, DETR

- Semantic & Instance Segmentation: FCN, U-Net, Mask R-CNN

Section 5 — Sequence Modeling & Recurrent Networks

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory (LSTM)
- Gated Recurrent Units (GRU)
- Bidirectional & Deep RNNs
- Sequence-to-Sequence Models & Attention
- Connectionist Temporal Classification (CTC)

Section 6 — Transformers & Attention

- Self-Attention Mechanism
- Multi-Head Attention
- Positional Encoding: sinusoidal, rotary (RoPE), learned
- The Transformer Architecture (Encoder, Decoder, Encoder-Decoder)
- Pre-training & Fine-tuning: BERT, GPT, T5
- Efficient Transformers: FlashAttention, Sparse Attention, Linear Attention
- Vision Transformers (ViT) & Cross-Modal Models

Section 7 — Generative Models

- Autoencoders & Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs): training, architecture, stability
- Normalizing Flows
- Diffusion Models: DDPM, DDIM, Score Matching
- Latent Diffusion & Stable Diffusion

Section 8 — Representation & Self-Supervised Learning

- Embeddings: word2vec, GloVe, FastText, contextual embeddings
- Contrastive Learning: SimCLR, MoCo, CLIP
- Masked Modeling: BERT-style, MAE
- Knowledge Distillation
- Transfer Learning & Domain Adaptation

Section 9 — Deep Reinforcement Learning

- RL Fundamentals: MDPs, policies, value functions
- Deep Q-Networks (DQN) and variants

- Policy Gradient Methods: REINFORCE, A3C, PPO
- Model-Based RL
- RLHF: RL from Human Feedback

Section 10 — Production, Efficiency & Advanced Topics

- Model Compression: pruning, quantization, knowledge distillation
- Hardware & Parallelism: data, model, tensor parallelism
- Inference Optimization: ONNX, TensorRT, KV-cache
- Interpretability & Explainability
- Fairness, Robustness & Adversarial Examples
- Scaling Laws & Emergent Behavior
- Mixture of Experts (MoE)



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